

270 2021 by Lin

1) $p = \frac{h}{\lambda}$, $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2m}v_0}$

2) $\hat{x} = x$, $\hat{p} = -i\hbar \frac{d}{dx}$

$[\hat{x}, \hat{p}]\psi = (\hat{x}\hat{p} - \hat{p}\hat{x})\psi = -i\hbar \frac{d}{dx}(x\psi) = -i\hbar x \frac{d\psi}{dx} + i\hbar \psi + i\hbar x \frac{d\psi}{dx}$

$[\hat{x}, \hat{p}] = i\hbar$

3) $-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + V(x)\psi(x) = E\psi(x)$

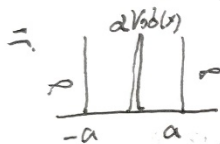
$x \leftrightarrow -x$

$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(-x) + V(-x)\psi(-x) = E\psi(-x)$, $\psi(-x)$ 也是方程的一个解 对应同一能量本征值 E

$\psi(x)$, $\psi(-x)$ 线性相关, 均为差一常数 C .

$\psi(x) = C\psi(-x) = C^2\psi(x)$

$C^2 = 1$ $\begin{cases} C=1 & \psi(x) = \psi(-x) \text{ 偶对称} \\ C=-1 & \psi(x) = -\psi(-x) \text{ 奇对称} \end{cases}$



1) 薛定谔方程: $-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + 2V_0 \delta(x) \psi(x) = E\psi(x)$

类似无限深势阱 $\psi(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$ $a \rightarrow 2a$ 左移 a

波函数 $\psi(x) = \sqrt{\frac{1}{a}} \sin \frac{n\pi(x+a)}{2a}$

2) 边界条件 $\psi(\pm a) = 0$

对原方程两边从 $x=0$ 到 $x=L$ 积分得阶跃条件

$-\frac{\hbar^2}{2m} \frac{d\psi}{dx} \Big|_{x=0}^{x=L} + 2V_0 \psi(0) = 0 \xrightarrow{L \rightarrow 0^+} \psi'(0^+) - \psi'(0^-) = \frac{2mV_0}{\hbar^2} \psi(0)$

3) 奇对称 $\psi(0) = 0$ $\sin \frac{n\pi}{2} = 0$ $n = 2, 4, 6, \dots$

$\psi_n(x) = \sqrt{\frac{1}{a}} \sin \frac{n\pi}{2a} (x+a)$ $n = 2, 4, 6, \dots$

$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2 \pi^2 n^2}{8ma^2}$ $n = 2, 4, 6, \dots$ 比无限深势阱高一半

4) 偶对称, 不为零 $x=0$ 处 $\frac{d^2}{dx^2} \psi(x) + k^2 \psi(x) = 0$ $k = \sqrt{\frac{2mE}{\hbar^2}}$

解 $\psi(x) = A \sin(kx + \varphi)$ $\psi(0^+) = \psi(0^-)$

边界条件 $\psi(a) = 0$ $A \sin(ka + \varphi) = 0$

$A \sin(-ka + \varphi) = 0$

阶跃条件: $KA \cos(ka + \varphi) - KA \cos(-ka + \varphi)$

4) 偶宇称, $\frac{d^2\psi}{dx^2} + k^2\psi = 0$

$$\psi(x) = A \sin(kx + \varphi)$$

$$\psi(\pm a) = 0 \quad \psi(x) = A \sin kx - ka \quad a < x < a$$

$$\psi(x) = A \sin kx + ka \quad -a < x < a$$

$\psi(0)$ 处连续

$$A \sin(ka) = A \sin ka \quad A = A$$

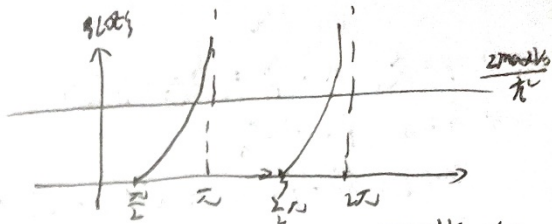
$$\psi(x) = \begin{cases} A \sin k(x-a) & x < -a \\ -A \sin k(x+a) & x > a \end{cases}$$

$$\psi'(0^+) - \psi'(0^-) = \frac{2ma\psi(0)}{\hbar^2}$$

$$Ak \cos ka + Ak \cos ka = -\frac{2ma\psi(0)}{\hbar^2} \sin ka$$

$$k \cot ka = -\frac{2ma\psi(0)}{\hbar^2} \quad \text{令 } \xi = ka$$

$$-\xi \cot \xi = \frac{2ma\psi(0)}{\hbar^2}$$



由图 定态能量由 $g(x)$ 与直线 $P = \frac{2ma\psi(0)}{\hbar^2}$ 确定

$$\xi_1 = ka = \sqrt{\frac{2mE}{\hbar^2}} a \quad E_1 = \frac{\xi_1^2 \hbar^2}{2am}$$

$$a \text{ 很小时 交点在 } (\frac{\pi}{2}, 0) \text{ 附近: } E_1 = \frac{\pi^2 \hbar^2}{8ma}$$

$$a \text{ 很大时 基态交点 } (\pi, 0) \quad E_1 = \frac{\pi^2 \hbar^2}{8ma}$$

$$\text{由 3) 奇宇称 能级最低 } E_1 = \frac{2^2 \pi^2 \hbar^2}{8ma} = \frac{\pi^2 \hbar^2}{2ma}$$

由此可见偶宇称 能级最低 E_1 才是基态。

$$1) \quad \hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}_+ + \hat{a}_-) \quad \hat{p} = i\sqrt{\frac{\hbar m\omega}{2}} (\hat{a}_+ - \hat{a}_-)$$

$$a_+ |n\rangle = \sqrt{n+1} |n+1\rangle \quad a_- |n\rangle = \sqrt{n} |n-1\rangle$$

$$\hat{x}|n\rangle = \sqrt{\frac{\hbar}{2m\omega}} (\sqrt{n+1} |n+1\rangle + \sqrt{n} |n-1\rangle) \quad \hat{p}|n\rangle = i\sqrt{\frac{\hbar m\omega}{2}} (\sqrt{n+1} |n+1\rangle - \sqrt{n} |n-1\rangle)$$

$$\langle x \rangle = \langle n | \hat{x} | n \rangle = 0$$

$$\langle p \rangle = \langle n | \hat{p} | n \rangle = 0$$

$$2) \quad \hat{x}^2 = \frac{\hbar}{2m\omega} (\hat{a}_+^2 + \hat{a}_-^2 + 2\hat{a}_+ \hat{a}_- + 1) \quad \hat{p}^2 = -\frac{\hbar m\omega}{2} (\hat{a}_+^2 + \hat{a}_-^2 - 2\hat{a}_+ \hat{a}_- - 1)$$

$$\langle x^2 \rangle = \langle n | \hat{x}^2 | n \rangle = \frac{\hbar}{2m\omega} (2n+1) \quad \langle p^2 \rangle = \frac{\hbar m\omega}{2} (2n+1)$$

$$3) \quad \Delta x = \langle (x - \bar{x})^2 \rangle = \langle x^2 \rangle - \langle x \rangle^2 = \frac{\hbar}{2m\omega} (2n+1)$$

$$\Delta p = \langle (p - \bar{p})^2 \rangle = \frac{\hbar m\omega}{2} (2n+1)$$

$$\Delta x \Delta p = \frac{\hbar^2}{4} (2n+1)^2 \gg \frac{\hbar^2}{4} \quad \text{当且仅当 } n=0 \text{ 时成立}$$

$$4) \quad \langle x \rangle = \langle \psi | \hat{x} | \psi \rangle = \frac{1}{\sqrt{2}} \langle a_1 \psi_1 + b_2 \psi_2 | \hat{x} | a_1 \psi_1 + b_2 \psi_2 \rangle$$

$$= (a_1 \langle 1 | + b_2 \langle 2 |) \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}_+ + \hat{a}_-) (a_1 |1\rangle + b_2 |2\rangle)$$

$$= (a_1 \langle 1 | + b_2 \langle 2 |) \sqrt{\frac{\hbar}{2m\omega}} (\sqrt{2} a_1 |2\rangle + \sqrt{3} b_2 |3\rangle + a_1 |0\rangle + b_2 |1\rangle)$$

$$5) \quad \psi = a \psi_1 \exp(-iEt/\hbar) + b \psi_2 \exp(-iEt/\hbar) \quad E_n = (n + \frac{1}{2}) \hbar \omega$$

$$\langle x \rangle = (a \langle 1 | e^{iEt/\hbar} + b \langle 2 | e^{iEt/\hbar}) \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}_+ + \hat{a}_-) (a |1\rangle e^{-iEt/\hbar} + b |2\rangle e^{-iEt/\hbar})$$

$$= ab \sqrt{\frac{\hbar}{m\omega}}$$

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四) 氢原子能量 $E_n = -\frac{e^2}{2an^2}$

可能值为 $E_1 = -\frac{e^2}{2a}$, $E_2 = -\frac{e^2}{8a}$ 相应概率为 $|A|^2$, $|B|^2 + |C|^2$.

2) L^2 本征值为 $\hbar^2 l(l+1)$, L_z 本征值为 $m\hbar$

可能值及相应概率: $L^2 = 0$, $|A|^2$, $L^2 = 2\hbar^2$, $|B|^2 + |C|^2$
 $L_z = 0$, $|A|^2 + |B|^2$, $L_z = \hbar$, $|C|^2$

3) ψ_{100} , $l=0$, $L_x=0$

ψ_{210} , ψ_{211} , $l=1$, $L_x = \hbar$ 或 0 . 设相应概率为 a , b , c

$$\psi(r, \theta, \phi) = AR_{10}Y_{00}e^{iEt/\hbar} + BR_{21}Y_{10}e^{iEt/\hbar} + CR_{21}Y_{11}e^{iEt/\hbar}$$

$$a+b+c=1 \quad a\hbar + c\hbar = 0 \Rightarrow a=c$$

$$\langle L^2 \rangle = \frac{1}{2} \langle L^2 - L_z^2 \rangle = \frac{1}{2} \langle L(L+1)\hbar^2 - m^2\hbar^2 \rangle$$

$$\begin{aligned} Y_{00}: l=1, m=0 \quad \langle L_x^2 \rangle &= \frac{1}{2} \langle L^2 - L_z^2 \rangle = \hbar^2 \\ Y_{11}: l=1, m=1 \quad \langle L_x^2 \rangle &= a\hbar^2 + c\hbar^2 = (a+c)\hbar^2 \end{aligned} \Rightarrow a+c=1$$

$$\Rightarrow a=\frac{1}{2}=c, \quad b=0$$

$$Y_{11}: l=1, m=1 \quad \langle L_x^2 \rangle = \frac{1}{2} \langle L^2 - L_z^2 \rangle = \frac{1}{2}\hbar^2$$

$$a+c=\frac{1}{2} \Rightarrow a=c=\frac{1}{4}, \quad b=0$$

$$\text{因此 } L_x=0 \quad |A|^2 = \frac{1}{4}, \quad L_x=\hbar \quad \frac{1}{4}(|B|^2 + |C|^2) \quad L_x=-\hbar \quad \frac{1}{4}(|B|^2 + |C|^2)$$

4) $\langle x \rangle = \int \psi^*(r, \theta, \phi) x \psi(r, \theta, \phi) d\tau$

$$= \int (A\psi_{100}e^{-iEt/\hbar} + B\psi_{210}e^{-iEt/\hbar} + C\psi_{211}e^{-iEt/\hbar})^* r \cos\theta \sin\theta \, r^2 \sin\theta \, dr \, d\theta \, d\phi$$

$$= 0$$

5) 可以. 因为 $\hat{H}$, $\hat{L}^2$, L_z 共同构成一个 C.S.O.

五. $\vec{B} = B\vec{e}_z$ m. q. $S_z = \frac{1}{2}$ $t=0$ $|0\rangle$

$$1) \hat{H} = -\vec{\mu}_B \cdot \vec{B} = \frac{eB}{\mu_C} S_x$$

$$S_x \text{ 本征值与本征态为 } S_x = \frac{\hbar}{2} \quad \psi_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad S_x = -\frac{\hbar}{2} \quad \psi_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad E_{\pm} = \pm \frac{eB\hbar}{2\mu_C}$$

t 时刻波函数 令 $\omega = \frac{eB}{\mu_C}$

$$\psi(x, t) = c_1 \psi_1 e^{-i\omega t} + c_2 \psi_2 e^{i\omega t} \quad \psi(x, 0) = c_1 \psi_1 + c_2 \psi_2 = |0\rangle$$

$$c_1 = \langle \psi_1 | \psi(x, 0) \rangle = \frac{1}{\sqrt{2}} \langle 1 | 1 \rangle = \frac{1}{\sqrt{2}} \quad c_2 = \langle \psi_2 | \psi(x, 0) \rangle = \frac{1}{\sqrt{2}}$$

$$\psi(x, t) = \frac{1}{\sqrt{2}} \psi_1 e^{-i\omega t} + \frac{1}{\sqrt{2}} \psi_2 e^{i\omega t} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-i\omega t} + \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{i\omega t} = \begin{pmatrix} \cos \omega t \\ -i \sin \omega t \end{pmatrix}$$

2) ψ 的本征值及本征态为

$$\psi_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \psi_- = -\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\psi_+ = \frac{1}{\sqrt{2}} \text{ 时 } P = |\psi_+^* \psi(x,t)|^2 = \left| \frac{1}{\sqrt{2}} (1+i) e^{-i\sin t} \right|^2 = \frac{1}{2} (1 + \sin t)^2$$

$$\psi_- = -\frac{1}{\sqrt{2}} \text{ 时 } P = |\psi_-^* \psi(x,t)|^2 = \left| \frac{1}{\sqrt{2}} (1-i) e^{-i\sin t} \right|^2 = \frac{1}{2} (1 - \sin t)^2$$

3) $\psi(x,t) = \frac{1}{\sqrt{2}} e^{i\sin t}$

第一次反方向时 $|\cos t| = 0$ $|\sin t| = 1$ 即 $\psi = \frac{1}{\sqrt{2}} (1+i)$ ~~$\frac{1}{\sqrt{2}} (1+i)$~~

$$\frac{eB}{2\mu c} = \frac{\hbar}{2\mu c} \sum_{n=1}^{\infty} T_n = \frac{\hbar^2 \mu c}{eB}$$

4) $t = \frac{\pi}{2}$ 时 $\psi(x, \frac{\pi}{2}) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ 再 $t = \pi$ 时

$$\psi(x, \pi) = c_1 \psi_+ e^{-i\pi} + c_2 \psi_- e^{-i\pi} \quad \psi(x, 0) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = c_1 \psi_+ + c_2 \psi_-$$

可能值 $\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}$

1) $\psi = 0$ ~~$\psi = 0$~~ $\psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$ $E_n = \frac{n^2 \hbar^2}{2ma^2}$

$$\psi_{11} = \psi_1(x_1) \psi_1(x_2) = \frac{2}{a} \sin \frac{\pi x_1}{a} \sin \frac{\pi x_2}{a} \quad E_1 = \frac{\pi^2 \hbar^2}{ma^2}$$

第一激发态 $\psi = \frac{1}{\sqrt{2}} (\psi_1(x_1) \psi_2(x_2) + \psi_2(x_1) \psi_1(x_2))$ $E = \frac{5\pi^2 \hbar^2}{2ma^2}$

2) $\psi = \frac{1}{\sqrt{2}}$

$$\psi = \frac{1}{\sqrt{2}} (\psi_1(x_1) \psi_2(x_2) + \psi_2(x_1) \psi_1(x_2))$$

$$\psi_1 = \psi_1(x_1) \psi_1(x_2) \quad E_1 = \frac{\pi^2 \hbar^2}{ma^2}$$

$$\psi_2 = \frac{1}{\sqrt{2}} (\psi_1(x_1) \psi_2(x_2) + \psi_2(x_1) \psi_1(x_2))$$

$$\psi_3 = \frac{1}{\sqrt{2}} (\psi_1(x_1) \psi_2(x_2) - \psi_2(x_1) \psi_1(x_2)) \quad E = \frac{5\pi^2 \hbar^2}{2ma^2}$$

7) 基态 - 一级修正

$$E_1^{(1)} = H_{11}' = \int \psi_1^*(x) \psi_1(x) (-aV_0 \delta(x-x_0)) \psi_1(x) \psi_1(x) dx$$

$$= -aV_0 \frac{4}{a^2} \sin^4 \frac{\pi x}{a} = -\frac{4V_0}{a} \sin^4 \frac{\pi x}{a}$$

第一激发态 - 一级修正

$$E_2^{(1)} = H_{22}' = \int \frac{1}{\sqrt{2}} (\psi_1(x) \psi_2(x) + \psi_2(x) \psi_1(x))^* (-aV_0 \delta(x-x_0)) \frac{1}{\sqrt{2}} (\psi_1(x) \psi_2(x) + \psi_2(x) \psi_1(x)) dx$$

$$= -\frac{aV_0}{2} \int (|\psi_1(x) \psi_2(x)|^2 + \psi_1^*(x) \psi_2^*(x) \psi_2(x) \psi_1(x) + |\psi_2(x) \psi_1(x)|^2 + \psi_2^*(x) \psi_1^*(x) \psi_1(x) \psi_2(x)) dx$$

$$= -\frac{aV_0}{2} \int |\psi_1(x) \psi_2(x)|^2 dx$$

$$= -\frac{8V_0}{a} \left(\sin \frac{\pi x}{a} \sin \frac{2\pi x}{a} \right)^2$$